Task 8 - TEA - Mandelbrot set or Julia's set

Instructions

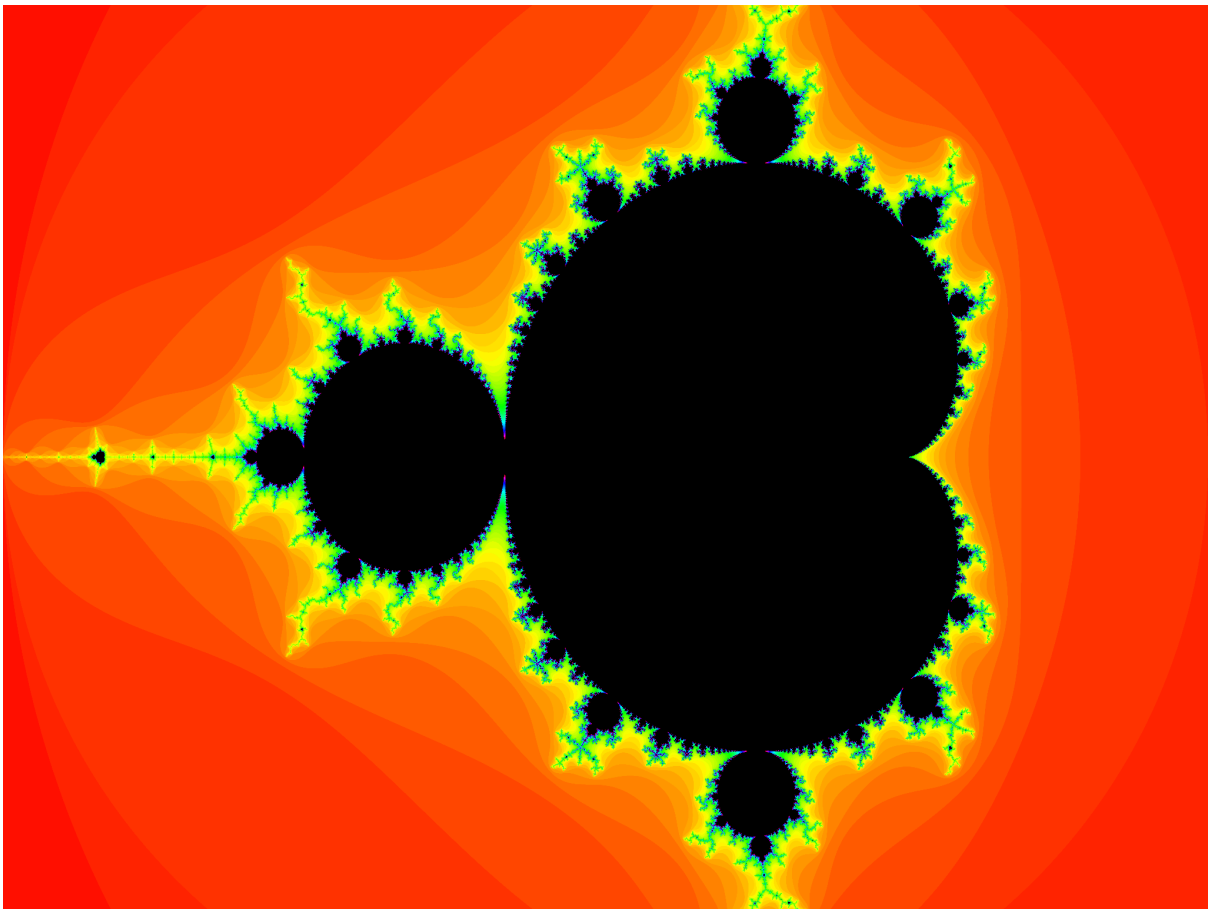
Implement the Mandelbrot or Julia set with zoom. It is defined as the set of complex numbers c, where the sequence z_0, z_1, ... is limited, i.e. it satisfies the following condition:

There exists a real number m such that for all n $|z_n| <= m$ (|.| is the absolute value)

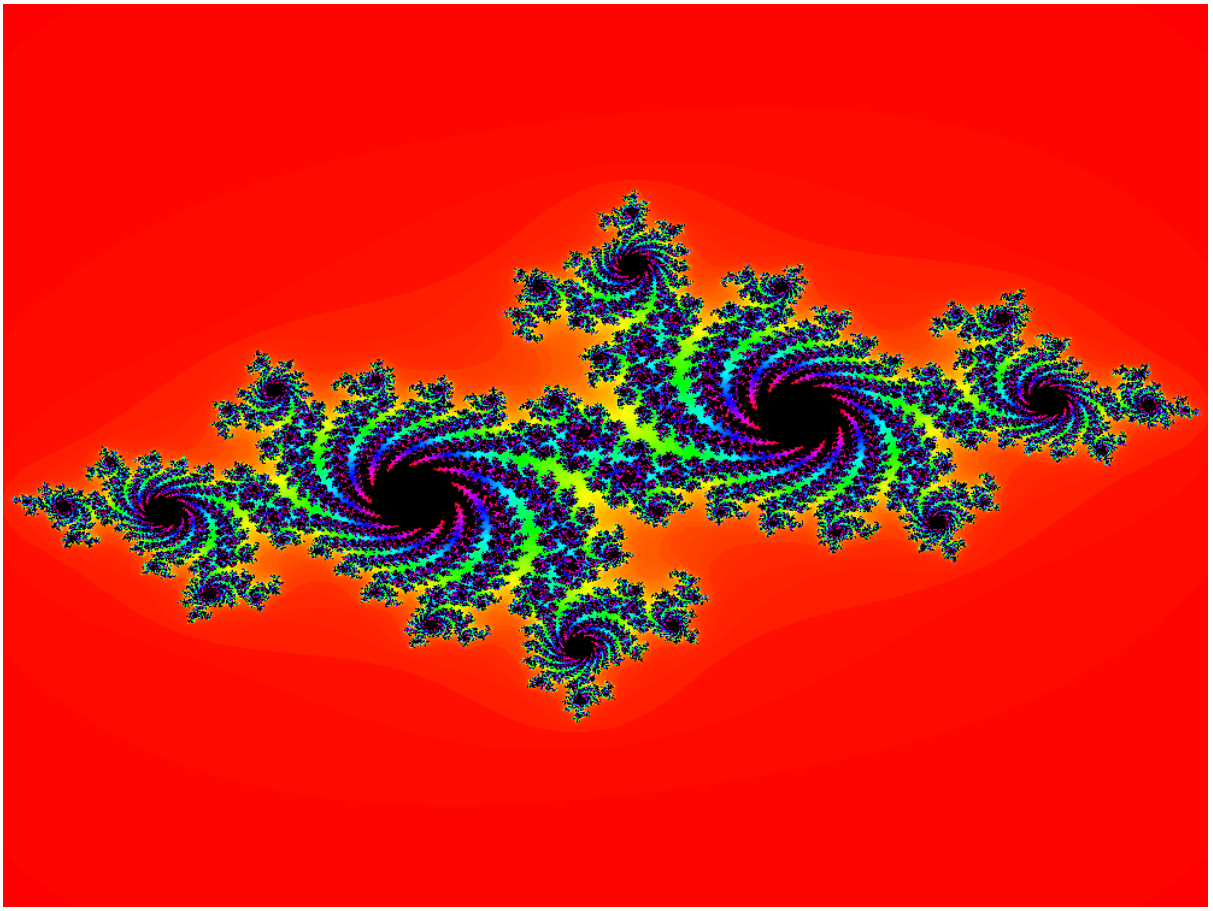
The sequence is generated according to the following equation:

$$z_0 = 0,\ z_{n+1} = z_n^2 + c$$

and $m = 2$. For a borders of the image you can use range of $X = \{-2, 1\}$; $Y = \{-1, 1\}$ for Mandelbrot set and $X = \{-1.5, -1.5\}$; $Y = \{-1.5, -1.5\}$ for Julia set. For setting a fill color of iteration nesting value a Hue color with saturation can be used.



Example 1: Mandelbrot set



Example 2: Julia set